Lab Report 01

Course title: Computer Network Laboratory
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Submitted to-

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Title: LAN Configuration and Connectivity Test Using Hub and Switch

Objective:
To configure IP addresses on PCs and test communication between them using a hub and a switch in a LAN.

Description:
First I took one hub and three PCs and connected them using wires. Then I opened each PC and went to Desktop - IP Configuration. I configured the IPv4 address for each PC. For the first PC, I entered 192.168.1.1 as the IPv4 address and the subnet mask was filled automatically after pressing Enter. I configured the IP addresses of the other two PCs in the same way. After configuring the PCs, I tested the connection using Message Simulation and the message was successfully delivered. I also used the ping command in Command Prompt to check the connection between the PCs and the ping was successful.

After that, I repeated the same process by replacing the hub with a switch. I connected the three PCs to the switch using wires and configured their IP addresses. Finally, I tested the connection using Simulation and the ping command. The connection worked successfully with the switch as well.

The main difference between the hub and the switch was observed during the message simulation. When using the hub, the message was sent to all connected PCs because the hub does not identify the specific destination PC. When using the switch, the message was sent only to the required destination PC. This shows that a switch provides more efficient communication than a hub.

Connection:

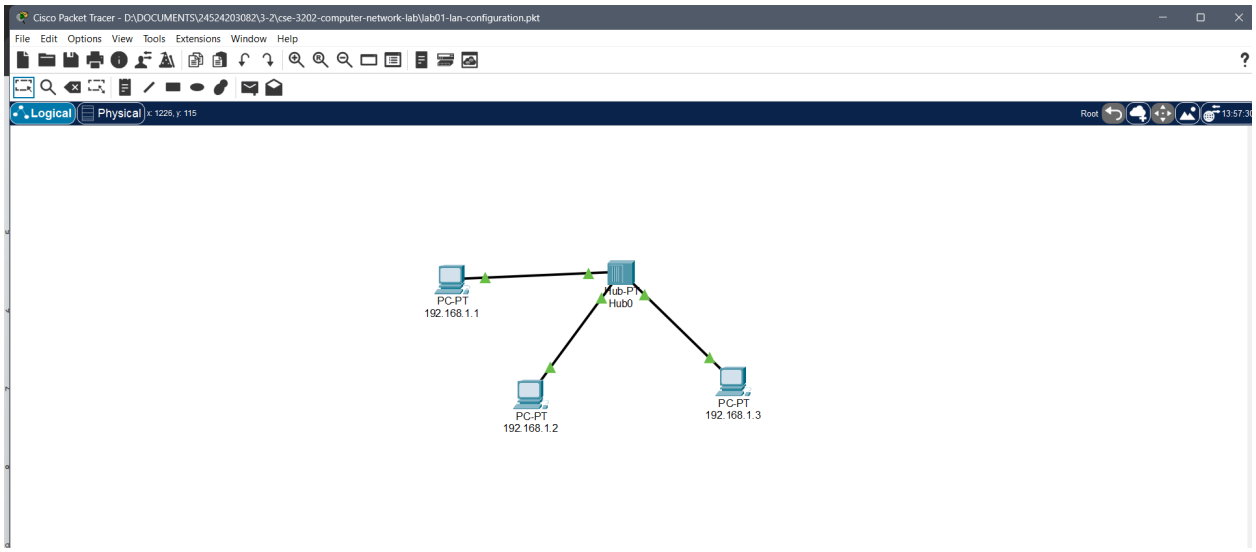


Figure 1: LAN Connection Using Hub

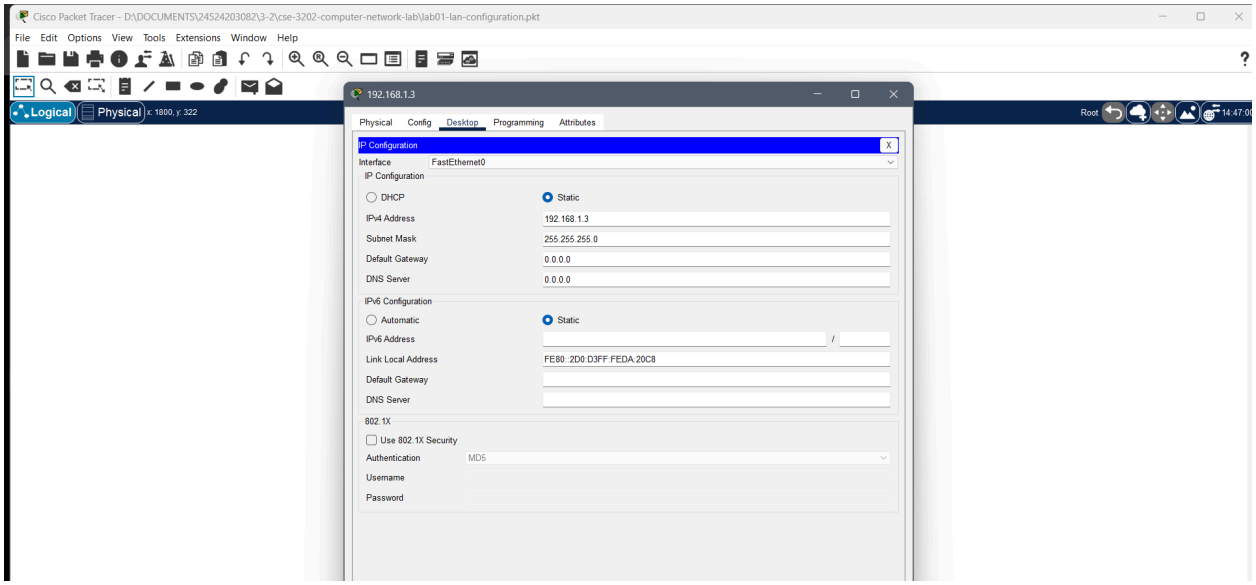


Figure 2: IP Configuration of PC

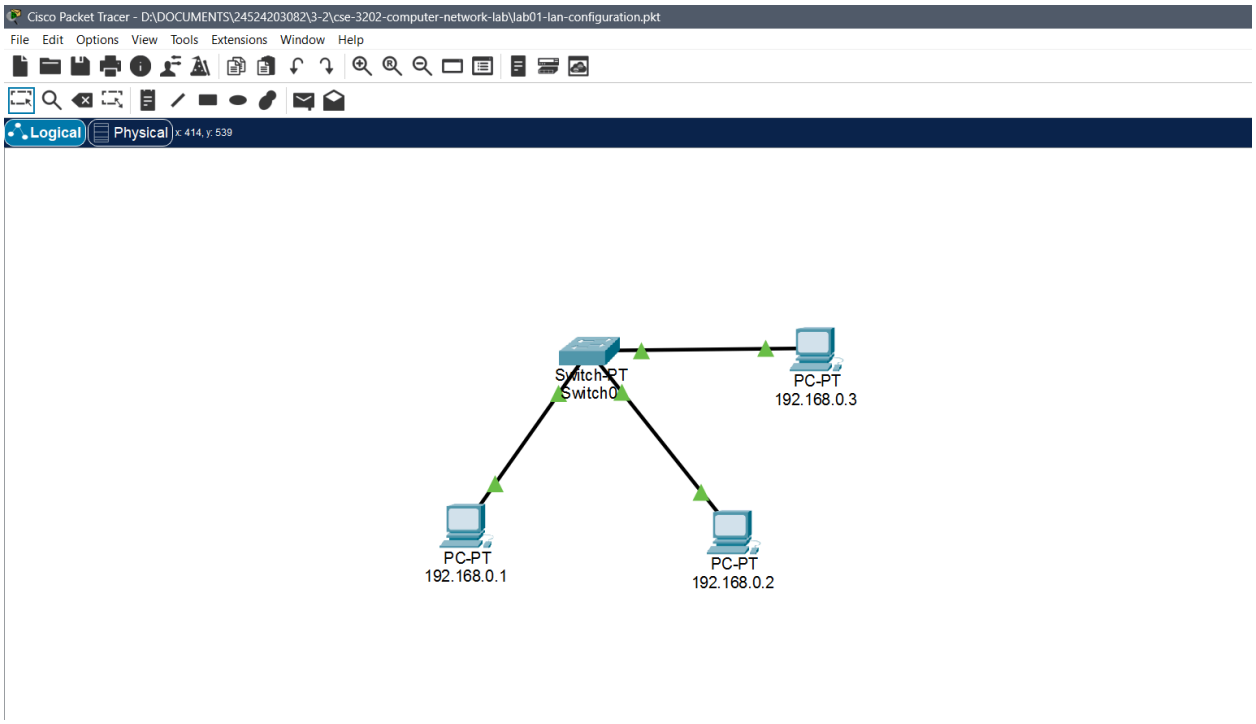


Figure 3: LAN Connection Using Switch

Command Prompt:

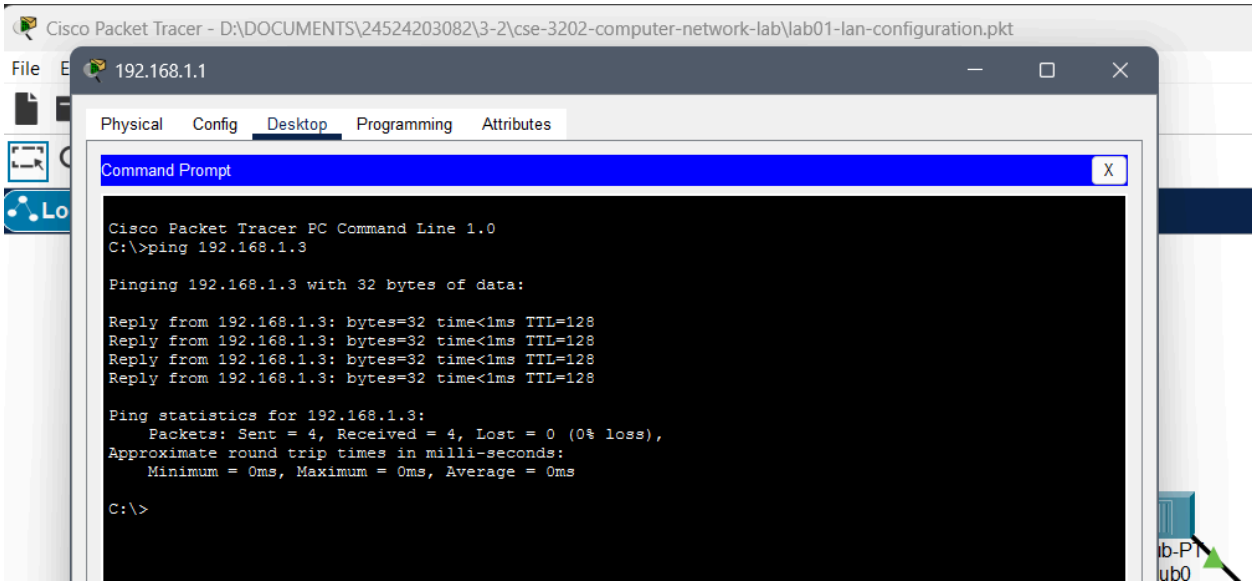


Figure 4: Ping Test (Hub)

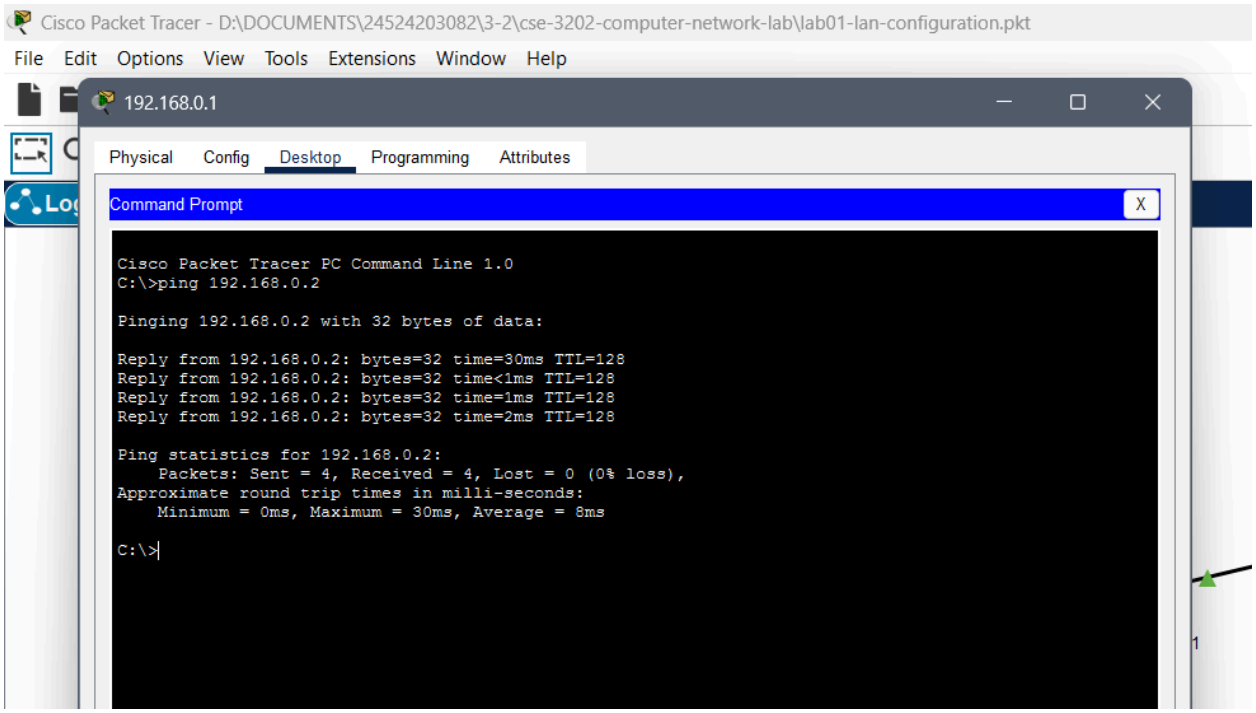


Figure 5: Ping Test (Switch)

Output:

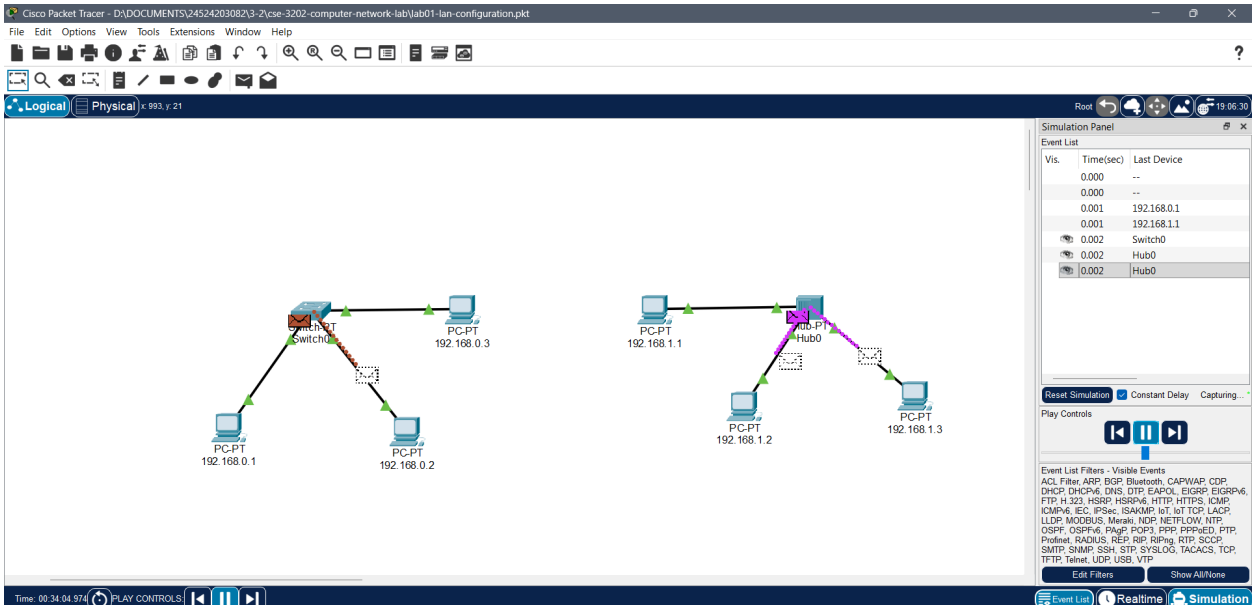


Figure 6: Message Simulation Output